Glossary

Industry Integrated AI System Synthesis: a weekly A&E demand forecast for Scottish hospitals, with demand briefs a duty manager approves

Patrick Gilliland

Every figure here is from the notebook's saved outputs or a source in the References below.

About this glossary

How it relates to the paper. The Reflective Synthesis Paper (`module_summary.pdf`) points to this glossary in one line under its References section: "Glossary: every term used in this paper is defined in glossary.pdf, submitted with it." That line is a pointer to my own file, not an APA reference. The glossary is not part of the paper's word count.

What it covers. Every term a reader or mentor could stumble on in the paper, the case study, the notebook and the README. Each term has one wording, used everywhere. Where a document once used another word for the same thing, the entry says so.

How to read it. Terms sit in six groups, alphabetical within each group; terms that start with a number come first. Each entry gives a plain-English definition, then "In this project:" with how the term is used here. Figures come from the executed notebook unless a source is cited, and every notebook figure is in-sample: the design was tuned on the same 2015 to 2026 data that scores it. Goal codes G1 to G8 are named at first use and in full in the entry "Goals G1 to G8".

Citations. The citations here resolve to the References at the end of this glossary, a short list taken from the case study's checked references.

Groups.

1. NHS and A&E
2. Data and timing
3. Forecasting and statistics
4. The design
5. Agents and governance
6. Project terms

1. NHS and A&E

4-hour standard

The Scottish Government's standard that 95% of A&E patients wait no longer than four hours from arrival to admission, discharge or transfer (Audit Scotland, 2025, p. 32). "4-hour performance" is the share who did.

In this project: the system never monitors the standard. Public Health Scotland says its weekly figures "should not be used to monitor against the 4-hour A&E access standard" (Public Health Scotland, 2026a, data set description). The share seen within 4 hours is lowest in December and January, while attendances peak in May and June, which is why the demand level is not a measure of pressure.

A&E (accident and emergency)

The UK name for a hospital's urgent care service for people who arrive without an appointment. It covers both emergency departments and minor injury units.

In this project: Scotland's A&E sites, minor injury units included, saw 137,703 attendances a month in 2025 (Public Health Scotland, 2026b), about 1.65m in the year.

A&E site

Any site counted in Public Health Scotland's A&E statistics: an emergency department or a minor injury unit.

In this project: "all A&E sites" (about 1.65m attendances in 2025) includes minor injury units, and the term is used only for that figure. The forecasts cover the 30 emergency departments (EDs) in the weekly data set, which saw 1,405,613 attendances in weeks ending in 2025, my sum from the weekly data set (Public Health Scotland, 2026a). The paper and case study call those 30 "emergency departments" (PHS's term), "EDs" or hospitals, never "A&E sites".

Audit Scotland

The independent public body that audits how Scotland's public money is spent.

In this project: its 2025 report gives the scale of supplementary staffing spend in 2024/25: £256m on agency staff of all kinds and £353m on the nursing bank (Audit Scotland, 2025, p. 15).

Bank and agency staff

Bank staff take extra shifts through the NHS's own staff bank, paid at standard NHS rates. Agency staff come from an outside agency and cost more; the extra cost is the agency premium.

In this project: NHS Scotland spent about £0.6bn in 2024/25 on agency staff of all kinds and the nursing bank: £256m and £353m. Medical and other bank spend is outside that sum. The two are never set side by side as like for like. Earlier notice could let cover come from the bank, not late from agencies, so any saving would mainly be the agency premium, not the whole spend. The wider value lies in staff welfare and patient care, links 3 and 4 of the value chain, which are sourced and hypothesised, not measured here. In 2025/26 agency nursing and midwifery fell to £34.5m while the nursing and midwifery bank rose to £391.0m, on hours up 0.1% (Public Services Delivery Scotland, 2026).

Duty manager

The manager on shift who runs a hospital's day-to-day operations, including staffing at the A&E front door.

In this project: the user of the system. The duty manager approves, edits or rejects each demand brief, and nothing is sent before approval. A brief marked "not verified, do not act" can never be approved. In the spot check the builder plays the approver, which is disclosed.

ED (emergency department)

A hospital's full A&E department for serious and urgent cases, as opposed to a minor injury unit.

In this project: "emergency department", or "ED" after first use, is PHS's term and mine for the 30 sites in the weekly data set, never "A&E site". The weekly data set holds 30 EDs with unbroken weeks. Scottish guidance says ED crowding "is linked to increased morbidity and mortality" (Scottish Government, 2016, Background section), which is why staffing the front door matters.

Front door

The point where patients arrive at A&E, and the staff who receive and assess them.

In this project: the harm addressed is a front door staffed for the usual week when demand is not usual. Beds and patient flow, which also drive waits, are out of scope.

Health board

One of the regional NHS bodies in Scotland that plans and runs health services, hospitals included, for its area.

In this project: "the board" is the body that would use the system. Its unscheduled-care lead owns the thresholds and its duty managers approve the briefs. A board using it for real would add its own beds and staffing data, which should stay inside its own network.

MIU (minor injury unit)

A smaller A&E unit that treats minor injuries, such as sprains and cuts, without the full services of an emergency department.

In this project: MIUs are inside the 1.65m attendances at all A&E sites, but outside the weekly data set the forecasts use.

NHS England

The body that runs the NHS in England.

In this project: two of its publications show the problem is real and owned: the OPEL escalation framework (NHS England, 2024) and the rota code (NHS England, 2025). Neither is Scottish; each is used for structure or realism only.

OPEL (operational pressures escalation levels)

NHS England's framework for escalation, in which a score built from set measures places a hospital at a level, and people decide the actions for each level.

In this project: the bands borrow OPEL's structure only, where a score sets the level and people decide the actions, not its 0 to 100 scale. Neither the 2016 Scottish ED guidance (Scottish Government, 2016) nor the October 2026 Scottish surge framework (Scottish Government, 2026) uses OPEL. In June 2025 the Scottish Government committed to an NHS Scotland OPEL system (Scottish Government, 2025, recommendation 33).

PHS (Public Health Scotland)

The national public health body that publishes NHS Scotland's health statistics, including weekly A&E open data.

In this project: the only health data source (Public Health Scotland, 2026a). The weekly A&E file, under the Open Government Licence, holds 608 week endings, w/e 22 Feb 2015 to w/e 11 Oct 2026, for 32 hospitals. The series is 30 hospitals and the national total over 593 weeks from w/e 7 Jun 2015, after the reconfiguration of Glasgow's emergency departments in May 2015. The newest three weeks are flagged provisional, because in my data test PHS revised its latest week or two slightly upward.

RCEM (Royal College of Emergency Medicine)

The UK professional body for emergency medicine.

In this project: RCEM (2026) estimates 818 deaths in Scotland in 2025 associated with long waits. It applies a ratio of one extra death for every 72 patients spending 8 to 12 hours in an ED to the 58,870 who waited 12 hours or more before admission to a ward. It is an estimate, never a count, and it is stated as the stakes, never as a claim. In its December 2025 survey one Scottish ED clinical lead reported decision fatigue and burnout among staff, and RCEM links crowding to burnout.

Rota and roster

A rota is the pattern of shifts a department needs covered. A duty roster names who works each shift.

In this project: in England, the rota code expects resident doctors in training to get duty rosters at least 6 weeks, and work schedules at least 8 weeks, before a new rotation starts (NHS England, 2025). So a 1 to 4 week forecast informs short-notice changes inside a rota already issued. The source is not Scottish.

Unplanned attendance

A visit to A&E that was not arranged in advance. In its weekly file Public Health Scotland counts these apart from planned attendances, mostly those booked after contact with a Flow Navigation Centre, which it adds to its "All" figure from w/e 8 Nov 2020.

In this project: the only measure forecast. "Attendances" in this project means unplanned attendances, except the two scale figures for 2025, about 1.65m at all A&E sites and 1,405,613 at the 30 EDs, which count all attendances.

Unscheduled care

NHS Scotland's term for care a patient did not book in advance, such as A&E and emergency admissions.

In this project: the board's unscheduled-care lead owns the thresholds, approves each latch reset and reviews the level's distribution by site. With the duty managers, the lead is the intended user at the weekly staffing review.

2. Data and timing

As of

A date that fixes what was known: a calculation made "as of" a date uses only data published by then.

In this project: every forecast, rule and agent tool takes the origin as its as-of date (`as_of`) and sees only weeks released by then. A unit test proves that no forecast uses a later week. It is a back-test on a revised snapshot: the 11 October 2026 file includes any revision PHS made after a week's first release, which a forecaster at the time would not have had. Without the cut-off, the March 2020 replay would see the collapse before it was published.

Data horizon

How many weeks after the last week of data a forecast is for.

In this project: a data horizon is weeks after the last week of data. Data horizon 1 is a finished week by the release. It labels only the data test's figures; the design speaks in release horizons, and release horizons 1 to 4 are data horizons 2 to 5.

Flag and keep

A cleaning approach that marks unusual records and keeps them, never dropping them silently.

In this project: taken from AI Programming Foundations. The weekly load flags duplicate keys, outlier weeks, the COVID period and provisional weeks, and drops no row: a row left out of the series keeps its reason. Keeping the COVID weeks lets the drift latch and the re-selection be tested on a real shock. This "flag" is a data mark, not the probability flag.

Hospital-week

One hospital's week.

In this project: the design counts in two fixed units, the outlook and the hospital-week counted once. The probe outputs used "hospital-week" for an outlook, one hospital at one origin, so a probe table's "hospital-weeks" may mean outlooks.

ISO week

The international standard numbering of weeks, Monday to Sunday, as weeks 1 to 52 or 53 of a year.

In this project: SA-MA's seasonal index is built by ISO week, and week 53 joins week 52.

OGL (Open Government Licence)

The UK licence that lets anyone reuse public-sector information, with attribution.

In this project: the PHS weekly A&E file and the GOV.UK bank-holidays file are both under OGL, so both ship with the repository.

Origin

The point in time a forecast is made from.

In this project: an origin is the latest week released at a Tuesday release, named by its week ending. The origin w/e 22 Mar 2020 was released on Tue 31 Mar 2020. An outlook is issued at one origin, and the replay set uses six fixed origins.

Publication lag

The delay between the end of a period and the publication of its data.

In this project: 9 days. Each Tuesday release covers the week that ended 9 days earlier, so the horizons are counted from the release, not from the last week of data.

Release

A publication of new data.

In this project: Public Health Scotland's weekly A&E file, published each Tuesday. A week is "released" once a release includes it, and every rule sees released weeks only.

Release horizon

How far ahead of the release a forecast week is.

In this project: a release horizon is one of the four week-endings after a Tuesday release, numbered 1 to 4. Release horizons 1 to 4 are data horizons 2 to 5. Release horizon 1 is the week already under way on release day.

Series

A run of values over time.

In this project: one hospital's weekly attendances, or Scotland's national total. There are 31 series, the 30 hospitals and the national series. The replay set uses five named series.

SHA-256

A fingerprint computed from a file's contents; any change to the file gives a different fingerprint.

In this project: the PHS file, `weekly_ae_activity_20261011.csv`, and the bank-holidays file are frozen by SHA-256, checked on every load and listed in the data manifest.

Target week

The week a forecast is for.

In this project: each target week has its own simulated capacity, and each week's own chance of being unusually busy is worked out for it.

Week ending (w/e)

The date of the Sunday that ends a reporting week, used to name the week.

In this project: weeks are named by their week ending, for example w/e 22 Mar 2020.

3. Forecasting and statistics

4-week moving average

A forecast equal to the average of the latest few weeks.

In this project: the 4-week moving average is the mean of the last four released weeks, used for every week ahead. It is never called a "moving mean" or a "rolling average". It is one of the four simple models, and the best simple model at every release horizon in the notebook's table for G1, forecasts a manager can plan with.

90th percentile

The value that 9 in 10 past values were at or below.

In this project: simulated capacity is the 90th percentile of the hospital's past weeks at the same time of year. The sensitivity table also shows the 80th and 95th percentiles.

Back-test

A test that replays past weeks as if live, makes each forecast from the data available then, and scores it against what happened.

In this project: the rolling-origin back-test replays every week as if live, each forecast using only weeks released by its origin, and scores every forecast. "Back-test" is its short form. Its settings are the data test's, fixed before any result, so they were not tuned to it.

Base rate

How often something happens anyway. In statistics it usually means the share of all cases. A warning means something only if it beats the base rate.

In this project: a base rate is always the share of unflagged cases, in the unit of the precision beside it. The outlook base rate is the share of unflagged outlooks that turned out busy, 21.2% in the steady years. The week base rate is the share of unflagged week-cells (one outlook's chance for one target week) whose week turned busy, 7.8% in the steady years. The busy share of all weeks, about 1 in 11, is never quoted as a base rate.

Best simple model

The better of the simplest benchmark forecasts, used as the bar a design has to meet.

In this project: the best simple model (G1) is the better of seasonal naive and the 4-week moving average at that release horizon, the two models that need no fitting, the bars to beat from Deep Learning Systems. Linear and SA-MA each estimate a seasonal shape or trend from the data, so neither is part of the bar.

Brier skill

A score for stated chances against a simpler benchmark. The Brier score is the average squared gap between each stated chance and what happened, counted as 1 if it happened and 0 if not. The Brier skill is how much lower it is than the benchmark's; 0 means no better.

In this project: in the steady years the flag's stated chance had a Brier skill of 0.128 against each hospital's real-time busy share, the share of its outlooks already settled at the origin that turned busy. That benchmark is not a base rate, which counts unflagged outlooks only.

Calibration

How closely stated chances match what happens: of the weeks given a 30% chance, about 30% should turn out busy.

In this project: each week's own stated chance runs about twice what happens. In the steady years it averaged 0.162, and 8.9% of those weeks were busy. So the brief quotes the hospital's track record, never the chance. The word is used only in this sense. Settling the design's settings and goals on the same data that scores them is design tuning, never "calibration".

Design tuning

Settling a design's settings and goals by trying them on data, before the final evaluation.

In this project: design tuning is what probes 1 to 6 did, on the same 2015 to 2026 data that scores the design. They settled the bands, the re-selection, SA-MA's two settings, the flag's 50% setting and four goals: G1, G3 (shocks are caught), G7 (raised levels are rare in steady years) and G8 (a flag tells a manager something). So every figure from it is in-sample, the choices made by a rule written first included. It is never called "calibration", which means only whether stated chances come true.

GRU (GRU comparator)

A gated recurrent unit: a small neural network, a model whose many internal weights are learned from examples, that reads a run of past weeks in order and carries a memory forward.

In this project: the GRU comparator is a small seeded GRU, run beside the simple models on the five named series, never eligible to set the flag and never shown to managers. It has one layer of 16 units, is refitted each January and averages three seeds. It comes from Deep Learning Systems. At the release horizons, outside each series' drift episodes, its paired interval favoured the primary model at 16 of 20 series-horizon pairs across the five replay series and the GRU at none. Outside March 2020 to March 2022 it favoured the GRU once, at the Queen Elizabeth University Hospital at release horizon 4. It stays a comparator: a yearly refit, three seeds and no one-sentence check for a manager.

In-sample

Measured on the same data that was used to choose the design or its settings, so likely to look better than on new data.

In this project: every figure in the notebook is in-sample: the design was tuned on the same 2015 to 2026 data that scores it. The back-test scores each forecast on weeks the model had not yet seen, but the design's settings were chosen on those same years. The notebook re-measures every figure, and shadow mode is the out-of-sample step.

Interval

A range likely to hold a true value (a confidence interval) or a future value (a prediction interval), given with its level, such as a 95% interval.

In this project: three kinds are kept apart. Every 95% interval in the evaluation comes from the moving-block bootstrap. Each brief shows an 80% and a 95% range per model and release horizon, from errors known at the origin. The drift interval is a third, separate interval used only for drift.

Lift

How many times more often a flagged case turns out busy than a case taken at random: the usual lift divides precision by the busy share of all cases.

In this project: the documents mostly say "times the base rate". Mine divides by the unflagged share, a risk ratio, which gives a larger figure than the usual lift. Outlook flags were borne out 2.54 times as often as the outlook base rate in the steady years, and G8 needs at least 2. The two units are never compared with each other.

Linear model

A forecast built from a straight-line trend, fitted by least squares, plus harmonics: pairs of smooth sine and cosine waves that repeat through the year and together trace its seasonal shape.

In this project: the linear model, "linear" for short, is the least-squares trend from Statistical Analysis plus four sine and cosine pairs on the week's place in the year (Hyndman & Athanasopoulos, 2021, Section 7.4), refitted at each origin on the trailing 156 weeks. It starts as the primary model for every series. From January 2018 to February 2020 it scored 2.71% nationally at data horizon 1, against 3.04% for the 4-week moving average and 4.17% for seasonal naive.

MAPE (mean absolute percentage error)

The average size of a forecast's miss as a share of the actual value, ignoring its direction.

In this project: MAPE, the mean absolute percentage error, is the average forecast miss as a share of the actual. On a slide it is "the average miss, as a share of the actual". In the notebook, nationally, from 2018 and outside the national drift episodes, the primary model scored 4.04% to 5.31% at the four release horizons, seasonal naive 9.19%, and SA-MA on its own 3.47% to 4.57% (Hyndman & Athanasopoulos, 2021, Section 5.8). Seasonal

naive's median percentage error, outside March 2020 to March 2022, is 11.0% at sites under 300 attendances a week, against 5.4% at 1,300 and over.

Mean steady year

An average year, used to put counts on a yearly scale.

In this project: the mean steady year is the mean of the four full steady years, 2019, 2023, 2024 and 2025. Figure 6 of the paper gives the range of its weeks given notice, 42 to 71. 2018 and 2026 are part years. Earlier drafts called it the typical steady year.

Moving-block bootstrap

The bootstrap re-draws the data at random, with replacement, many times to see how much a figure would vary. The moving-block version re-draws runs of neighbouring weeks together, because errors in nearby weeks are related.

In this project: the moving-block bootstrap resamples in 8-week blocks, because weekly errors are not independent, with 2,000 resamples. It gives every 95% interval in the evaluation.

No measurable gain

The plain reading of an interval for a difference that includes zero: the data cannot tell the two apart.

In this project: the notebook says "no measurable gain" at a horizon whose interval spans zero, and only there. The primary model against the 4-week moving average showed no measurable difference at release horizons 1 to 3 (-0.18, -0.37 and -0.55 points, each interval crossing zero). At horizon 4 it was 0.74 points lower (-1.22 to -0.18).

Non-shock weeks

The weeks left after taking out the periods when the forecasts had stopped fitting.

In this project: for G1 and G2 (honest uncertainty), the target weeks outside the national drift episodes, first breach to last alarm. It is a scoring rule only: no week is left out of the forecasting models' training, though SA-MA's seasonal index leaves out latched weeks.

Out-of-sample

Measured on data that played no part in choosing the design or its settings.

In this project: not yet done. The live out-of-sample run is shadow mode on new weekly releases, a pre-deploy step that is not done before submission.

Paired difference

The difference between two models' errors on the same weeks, which removes the effect of easy and hard weeks.

In this project: the notebook reports the GRU's paired difference against the primary model at each release horizon, with a 95% interval. The method comes from Statistical Analysis.

Point forecast

A single best-guess number for a week, without its range.

In this project: each simple model gives a point forecast with its ranges. The point forecast's own band is shown in the forecast table, never as the level.

Precision

The share of warnings that turned out right.

In this project: there are two, one per unit, each quoted beside its own base rate. In the steady years 53.8% of outlook flags were borne out, against an outlook base rate of 21.2%. 27.6% of week flags were right, against a week base rate of 7.8%. The two precisions are never compared with each other.

Range

The spread of values a forecast expects for a week.

In this project: each brief shows an 80% and a 95% range per model and release horizon, from the errors known at the origin. G2 scores the 95% range. Ranges are kept apart from the drift interval.

Recall

The share of real events that were warned of.

In this project: recall is the share of unlatched outlooks that turned out busy which were flagged, per outlook. On a slide it is "how many busy outlooks it flagged". It was 21.0% in the steady years. It is never quoted as a share of busy weeks: that is "given notice", which counts each hospital-week once.

Reliability table

A table that groups stated chances into bins and compares each bin's average chance with how often the event happened.

In this project: the notebook's reliability table shows the flag's stated chance against what happened, which is the evidence that the stated chance runs high.

Rolling origin

A testing method that treats each week in turn as "now" and forecasts only from what was known then.

In this project: the back-test uses a rolling origin every week (Hyndman & Athanasopoulos, 2021, Section 5.10). It extends the time-based hold-out from Statistical Analysis.

SA-MA (seasonally adjusted moving average)

A moving average that first takes out the time-of-year pattern, then puts back the pattern for the week being forecast.

In this project: SA-MA, the seasonally adjusted moving average, is the mean of the latest four released weeks, each divided by its own week's seasonal index, times the target week's index. The index is built by ISO week from the latest three complete calendar years, reaching back up to 8 years past latched weeks; both choices are in-sample. It joined the re-selection pool at my request, and on its own it had the lowest national error in the notebook's G1 table, 3.47% to 4.57%. The paper calls it my own seasonally adjusted moving average.

Seasonal index

A set of factors, one per week of the year, showing how busy that week usually is against the year's average, where 1 is average.

In this project: SA-MA's index, built for each hospital at each origin from released weeks only. Each week of the latest three complete calendar years is divided by its year's mean, the ratios are averaged by ISO week, smoothed across neighbouring weeks and scaled to average 1. Latched weeks are left out of the index only.

Seasonal naive

A forecast that repeats the value from the same time last year.

In this project: seasonal naive is the same week last year. It is never shortened to "naive", which would mean last week's value. It is the weakest model at the release horizons in the notebook, at 9.19% nationally, because that window keeps 2020 to 2022, when it copies COVID weeks. Outside March 2020 to March 2022 my data test found it 3.77% at every horizon.

Sensitivity table

A table that shows how the results change when a setting changes.

In this project: the share of hospital outlooks at each demand level, with capacity at the 80th, 90th and 95th percentiles. It shows what a different setting would cost, and that the level can rise.

Standard deviation

A measure of spread: how far values typically sit from their average.

In this project: the drift interval is ± 1.96 standard deviations of the primary model's last 52 one-week errors.

Steady years

Years free of a major shock, used to judge normal running.

In this project: the steady years are 2018 (from July), 2019 and 2023 to 2026, the years G7 and G8 score. G3 calls the same six calendar years "normal years", with all of 2018. 2020 to 2022 are left out because of COVID.

4. The design

Bands

Ranges of a measure that map to levels.

In this project: the level's bands come from the ratio of attendances to simulated capacity: level 1 below 1.05; 2 from 1.05 to under 1.10; 3 from 1.10 to under 1.15; 4 at 1.15 and above. Each band includes its lower edge. They grade the actual weeks (busy is level 2) and the surge level. The bands are my own design on OPEL's structure.

Bias by hospital size

Unfairness that falls on hospitals by their size.

In this project: seasonal naive's median error is 11.0% at sites under 300 attendances a week and 5.4% at 1,300 and over. The raised share varies widely by site: the five most raised, two children's hospitals and three hospitals under 800 attendances a week, raised 35.7% to 18.8% of their outlooks, while Glasgow Royal Infirmary, a large adult ED, raised none. Harm runs both ways, so every brief shows its own site's error and track record, and the level is never read as need.

Borne out

Shown by events to have been right.

In this project: borne out describes an outlook flag whose outlook, as it turned out, had at least one week at level 2 (1.05 times simulated capacity). It is used only of outlook flags, always beside the outlook base rate. A week flag is "right" when its named week turned busy.

Busy (unusually busy)

Demand well above what is usual for the time of year.

In this project: a hospital-week is unusually busy when its attendances reach 1.05 times its simulated capacity, that is level 2. An outlook is busy when any of its four weeks is. In the mean steady year about 142.5 of 1,560 hospital-weeks, 1 in 11, are unusually busy.

Capacity (simulated)

The number of patients a department can handle in a week. Real capacity depends on beds, staff and flow.

In this project: simulated capacity is, for a target week, the 90th percentile of the hospital's weekly attendances in the same 13-week window in each of the previous three years. The window is centred on that week and uses released weeks only. On first use it is "simulated capacity, a seasonal baseline of past demand". It is never beds or staff, so every level is illustrative.

Coordinator

The part of the system that turns an approved brief into tracked actions.

In this project: a small state machine in code, a fixed set of steps the code moves through. It assigns the front-door staffing actions to roles, writes notifications to an outbox file that is never sent, tracks simulated acknowledgements and escalates after a deadline. Code sets each action's text; the language model drafts only each notification's wording, which may state only figures the approved brief states. Every decision is recorded.

Demand brief

A short written briefing for a manager about expected demand.

In this project: the demand brief is the agent's output for one hospital and one week, checked by code and approved by a person. It is never called an "escalation brief" or a "report". Every brief prints the level and its meaning, the level's source, the hospital's own back-test error and track record, and the range, or why no range can be trusted while the drift latch is set. It also prints model agreement, the drift alarm (its field is `drift_alarm`), watch points, and the backend and model that wrote it. A brief is a demand-week, surge or latch brief by its level's source; the two demand-week briefs in the replay set are the ones a forecast flag raised.

Demand level

A graded measure of how high demand is.

In this project: the demand level shows demand unusually high for the time of year, for front-door staffing. On an unlatched week it is raised to level 2 when the flag is set, and is 1 otherwise. A surge reads levels 1 to 4 from the latest released actual against simulated capacity, and a held week keeps the flag of the week before its alarm. It is never named for pressure or escalation, and never read as need or priority. Every brief prints: "The demand level shows demand unusually high for the time of year. It is not pressure, need or priority."

Drift

The point at which a model's forecasts stop fitting what is happening.

In this project: drift is checked on the primary model's one-week-ahead forecasts against the drift interval. "Alarm" is used only for drift, never for the flag.

Drift alarm

A signal that the forecasts have stopped fitting.

In this project: the alarm fires when the actual falls outside the drift interval for two released weeks running, and it sets the drift latch. In March 2020 the first breach was w/e 15 Mar, and the alarm fired at w/e 22 Mar, known on Tue 31 Mar 2020. It confirms a shock; it does not warn of one. On the brief its field is `drift_alarm`. It is never called a "drift flag", because "flag" is kept for the probability flag.

Drift interval

A range around a forecast that the actual should fall inside if the model still fits.

In this project: the drift interval is the primary model's one-week-ahead forecast ± 1.96 standard deviations of its last 52 one-week errors, with at least 26 needed. It is kept apart from the brief's 80% and 95% ranges. Its basis, a forecast plus or minus 1.96 standard deviations, is a standard prediction interval (Hyndman & Athanasopoulos, 2021, Section 5.5); the two-week alarm and the latch are my own design.

Drift latch

A switch that stays on once set, until someone resets it.

In this project: the drift latch is set when the actual falls outside the drift interval for two released weeks running, like an alarm that stays on until someone resets it. While it is set, the series' forecasts are marked untrusted, the hospital is screened in, and only a person can reset it. It is never called an "untrusted latch". "Unlatched" means the latch is not set.

Escalation

Raising a matter to a more senior person or a higher level of response.

In this project: escalation is the coordinator's chase of a staffing action not acknowledged by its deadline. It is never the demand level. In the system's name it means that chase; in OPEL and Scotland's guidance it keeps their own meaning, quoted as theirs.

Extra patients

Attendances above a line.

In this project: extra patients are attendances above the hospital's simulated capacity in a busy hospital-week counted once. They are a scale, never a harm or a cost. In the mean steady year the weeks given notice held about 4,100 extra patients.

Flag

A mark that draws a manager's attention to a week.

In this project: the flag, or probability flag, is set on a week when its own chance of being unusually busy (reaching level 2) is at least 50%, at any of the four release horizons. An unlatched outlook with a flagged week carries the flag. It is not the chance that any of the four weeks is busy. It is never called an "alert" or "alarm", which is drift. The 50% setting was chosen by a rule written before its results, on the same data, so it is in-sample. In the steady years 9.5% of the unlatched outlooks whose weeks are known carried a flag.

Given notice

Warned of in advance.

In this project: a busy hospital-week counted once is given notice when at least one flagged unlatched outlook covered it, whichever of its four weeks carried the flag. In the steady years 280 of 713 busy hospital-weeks had notice; by the week's own flag it was 197 of 713. A week covered only by latched outlooks is out of scope in both counts. "Before the week began" applies when that outlook was released before the week began, at release horizons 2 to 4, six days or more ahead. It is never "a week or more of notice" or "a week or more ahead". In the mean steady year, 57.0 of 142.5 busy hospital-weeks had notice, 49.0 of them before the week began.

Graduated response

A light action on a first warning and a firmer one when it repeats.

In this project: tested and dropped. A repeated week flag was right 29.2% of the time in the steady years, against 25.6% for a first one, both against a week base rate of 7.8%. The intervals overlapped, and a repeat came about a week later.

Held level

A level kept from an earlier week when the latest forecasts cannot be trusted.

In this project: after a downward drift breach, a held week keeps what the flag said in the week before the alarm, the last trusted week, and the forecast is marked untrusted.

Hospital-week counted once

A unit for counting real weeks without double counting.

In this project: one hospital's week as it happened, counted once however many outlooks covered it. It is unusually busy when its attendances reach 1.05 times its simulated capacity. Totals of busy weeks, weeks given notice and extra patients use it.

Latch reset

Turning a set latch off.

In this project: in use, a latch reset would be the unscheduled-care lead's approval, logged, after the code's review prompt at 8 clear released weeks, and the analytics lead would re-select the primary model at each reset. A reset should never be automatic, because it restores trust in a hospital's forecasts. In the spot check, code resets the latch at once at its prompt and re-selects the model, a simulated step that stands in for that approval.

Option A

A rule for judging whether one text adds value over another.

In this project: option A is the template comparison's rule, fixed before the ratings: the agent's text adds value in a week when it meets every criterion the template's text meets and at least one more, with no factual error in it.

Outlook

A short-range view of the weeks ahead.

In this project: an outlook is one hospital's four-week forecast issued at one origin, covering release horizons 1 to 4. It is busy when any of its four weeks reaches level 2. G8's flag rate, borne-out share, base rate and recall, "flags raised" and G7's raised share count outlooks.

Outlook flag

The flag read for a whole outlook.

In this project: an outlook flag means an unusually busy week is likely somewhere in the next four. Its precision is the share borne out, quoted beside the outlook base rate: 53.8% against 21.2% in the steady years, about half against about 1 in 5. It is G8's unit. The mean steady year raised 145 outlook flags, 66 of them not borne out.

Pressure

How stretched a hospital is: patients waiting against beds, staff and flow.

In this project: the level is never named for pressure. The open data holds no beds, staffing or occupancy. In the steady years 29.9% of raised outlooks fell in that hospital's worst quarter of 4-hour weeks, against 25.4% by chance: a weak link, not a pressure measure.

Primary model

The model whose forecast a system acts on, when several are run.

In this project: the primary model is the simple model whose forecasts set a series' flag and drift check. Linear at the start; re-selected for that series at each latch reset among seasonal naive, the 4-week moving average, linear and SA-MA. The GRU is never eligible.

Raised

Set at a level above normal.

In this project: an outlook is raised when it is at level 2 or above, latched or not, from the flag, a surge or a held level. G7 is the share of outlooks raised, by the origin's calendar year. Probe 7 calls the same unit a hospital-week.

Re-selection

Choosing the model again in the light of recent performance.

In this project: at each latch reset the primary model is re-selected for that series among the four simple models, by MAPE at release horizon 2 on the latest 104 released weeks. The new primary is used from the next origin. In use the analytics lead runs it; in the spot check code runs it at each simulated reset. In the notebook the primary switched 55 times across the 31 series.

Replay set

A fixed set of past weeks on which a system is run to show its behaviour.

In this project: the replay set is the five named series at the six fixed origins, which give seven demand briefs (4 latch, 1 surge, 2 demand). The series are the national series, the Royal Infirmary of Edinburgh, Ninewells, Lorn & Islands and the Queen Elizabeth University Hospital. The origins run from w/e 6 Oct 2019 to w/e 9 Apr 2023, and were fixed by rules written before any replay: five named in the design probes, the sixth found by a stated rule.

Review prompt

A reminder that a latch may be ready to reset.

In this project: each released week without an alarm counts as clear, and a new alarm restarts the count. After 8 clear weeks the code raises a review prompt. In use it would wait in the approval queue for the unscheduled-care lead; in the spot check the reset follows at once, a simulated step.

Screen

The step that decides which cases go forward.

In this project: code picks the hospitals that are flagged, at a surge or held level of 2 or more, or with the drift latch set. Nothing else reaches the agents. The screen sits in the code lane of the paper's Figure 1.

September residue

What is left of an earlier quirk after a fix.

In this project: in the steady years 12.8% of September outlooks are flagged, and 40.4% of those flags are borne out, against 14.9% of unflagged September outlooks. An earlier design bunched its flags in September; under the approved design February has the highest flag rate. My reading is that the 4-week moving average's lag, where it is still the primary, causes part of the rest. The probes did not test that.

Staffing actions

The short-notice changes a department can make to cover a busy week.

In this project: front-door staffing actions only, set by code for the level and assigned after approval. Level 1 sets none. Level 2 asks the duty manager to check whether extra shifts or bank staff are needed and the rota coordinator to hold new leave requests; level 3 adds a request to the staff bank; level 4 also tells the unscheduled-care lead. The actions, owners and reply deadlines are illustrative.

Stated chance

The probability a system states that a given week turns out unusually busy.

In this project: a week's own stated chance, the flag's input, is the share of the primary model's last 104 released errors at that horizon, at least 26, which applied to its forecast reach 1.05 times capacity. It is not the chance that any of the four weeks is busy. In the steady years it averaged 0.162, and 8.9% of those weeks were busy, so no brief quotes it, and the brief has no field for it.

Surge

A sudden rise in demand.

In this project: a surge is a drift latch set by an upward breach. The level then comes from the latest released actual, and the brief is marked "surge: forecasts untrusted".

Template brief

A fixed fill-in-the-blanks text.

In this project: the template brief is the fixed fill-in-the-blanks text that code fills from the same tool results the Investigator uses, with no free text. "The template" is used only inside the template comparison. Its fields were fixed before any brief existed: the level and its meaning line, the four simple models' forecasts with ranges, the hospital's back-test error and track record, the drift status and one fixed watch point. It never ranks the models. "The APA template" is a different thing, always named in full.

Template comparison

A test of what the language model adds, set against a plain template.

In this project: the template comparison sets the template brief and the agent's demand brief for the same week side by side. It runs in two weeks chosen by rule: the national latch week, w/e 22 Mar 2020, and Ninewells' demand week, w/e 13 Jun 2021. Reader B, a person, rates them against the approval criteria with an AI assistant and the order hidden. The approver also rates them with an AI assistant; those ratings are not blind, as the approver had read every agent brief and was told the order. Neither pair was blind to Reader B, who had read both agent texts as briefs, the demand week's pair is not blind because the agent's text carries its NOT VERIFIED banner, and the template was Text A, shown first, in both weeks.

By option A, the rule fixed before the ratings, the agent adds value in a week only if its text meets every criterion the template's meets and at least one more, with no factual error. In the national latch week it did, for both raters, on the one criterion the template cannot meet: it addressed the models' agreement, saying every forecast was untrusted while the drift latch was set and ranking none. In the demand week it did not, for either rater: its text was the refused demand-week brief, whose trend and fit claims the checks had already stopped. So it added value overall for neither rater.

By the same rule, a demand-week brief now takes the template's explanation by default, and the agent writes the explanation in latch, surge and model-disagreement weeks; the language model still drafts each notification's wording. No threshold for model disagreement was set before the ratings, so that ground keeps no week for the agent; setting one is a next step. The template's no-alarm drift line, which both raters marked a factual error, is versioned. The comparison stays on the rated version 1; the shipped template and new runs use the raters' wording, "No drift alarm is recorded; this alone does not establish forecast fit." The recorded runs replay unchanged, and shadow mode tests the default.

Text A

The text a rater reads first in a pair.

In this project: in the template comparison each pair is shown as Text A and Text B, in a seeded order kept from the raters until they finished. The order put the template as Text A in both weeks.

Track record

A record of how often past warnings came true.

In this project: the track record is how often a hospital's past flags were borne out, with their count, from outlooks whose weeks were all released by the origin. It sits beside how often the hospital's unflagged outlooks turned busy. The brief quotes it, never the flag's stated chance, which runs high.

Watch point

A line telling a manager what to check.

In this project: each names something a manager can check or do this week. The template brief uses one fixed sentence chosen by rule; the agent writes its own, which approval criterion 5 judges.

Week flag

The flag read for one named week.

In this project: a week flag is one named target week whose own chance reaches 50% on an unlatched outlook, the week a brief would name. Its precision is the share whose named week turned busy ("right"), quoted beside the week base rate: 27.6% against 7.8% in the steady years, about 1 in 4 against about 1 in 13. It is never set beside an outlook figure.

5. Agents and governance

Adequacy test

A test, stated before any brief existed, of whether briefs are good enough to send for approval.

In this project: a brief is adequate when it is verified and meets at least 5 of the 6 approval criteria, as an AI reviewer, not the system's Reviewer, scores it, never the approver. A set needs 6 of its 7. The local model verified none in either round, so the frontier model ran once on that stated case. It verified 5 of 7, short of the 6, and the approver chose its set on those facts.

Agent

A language model given a goal and tools, which decides which tools to call and writes its output from their results.

In this project: one agent, the Investigator, and one language model check, the Reviewer's wording check. They explain; they never set a level or supply a figure of their own. Figure 1 and the paper call the Investigator and the Reviewer together "the agents".

AI independent judge

An AI model asked for an independent reading before a person decides.

In this project: the AI independent judge read the seven frontier briefs before the approver decided. The approver adopted its five edits, none of them factual, with the reset wording corrected, and its notes on the two rejections. It marked itself as advice, never the approver's sign-off.

AI reviewer

An AI model that scores work against stated criteria.

In this project: the AI reviewer scored the adequacy test. It is not the system's Reviewer, so the approver's approval stayed separate from it.

Alert fatigue

The habit of ignoring warnings when too many turn out false.

In this project: the mean steady year raised 145 outlook flags, 66 of them not borne out, and drift alarms and latched weeks screen hospitals in too. The controls are the two-week drift rule, the 8-week review prompt, the level's meaning and the track record on every brief, and a person approving each one.

Analytics lead

The analyst who looks after a board's forecasting models.

In this project: runs the re-selection at each latch reset and owns any other model change, each act logged. In the spot check the builder plays it.

Approval criteria

The tests a person applies before approving.

In this project: the approval criteria are the six criteria written before any brief existed, used by the approver, the second reader and the template comparison:

1. Figures correct.
2. Uncertainty stated: the range, the hospital's back-test error and its track record.
3. Model disagreement named, with which forecast to trust less and why.
4. The level's meaning stated.
5. Watch points actionable.

6. No clinical instruction.

Approval queue

The list of items waiting for a person's decision.

In this project: demand briefs wait in an approval queue file until the duty manager approves, edits or rejects each one, with initials and a timestamp. In use, latch review prompts would wait there too; in the spot check they are simulated. The pattern comes from Agentic AI Systems.

Approver

The person who approves, edits or rejects a system's output before it takes effect.

In this project: the approver is the duty manager's role at the approval queue. A person can disregard, override or reverse every output, following the principle in European Union (2024, Article 14), cited as a principle, not a legal duty. In the spot check the builder played the approver, which is disclosed and is the limitation the second reader exists to check. Documents name the role, never a person.

Audit log

A record of every step and decision, kept so that it can be checked later.

In this project: every tool call, draft, check and decision is logged, with the backend that wrote each brief, and can be replayed. In use, latch resets and re-selections would be logged with initials and a time; in the spot check they are simulated steps.

Automation bias

The tendency to accept a computer's output without enough checking (Goddard et al., 2012).

In this project: the main ethical risk. It is met by what every brief shows: what the level means, its range or why no range can be trusted while the drift latch is set, the models' agreement and the hospital's track record where it has one. It is also met by a person approving, and by the second reader in the evaluation.

Clinical safety reviewer

The person who confirms a system is safe before clinical use.

In this project: signs off before any live use and holds the open risk on the medical device question. The sign-off is not exercised, because there is no live use.

Code figure check

An automatic check, written in code, that every number in a text matches its source.

In this project: the first stage of the Reviewer. Every number in the brief's text must map to a cited figure, each cited figure must equal a tool result within rounding for the same hospital, week, model and horizon, and the level must equal the rule's. It is never described as "the language model checks the figures".

Factual edit

An edit that corrects a fact.

In this project: a factual edit is an edit that changes a figure, a date, a hospital or the level, or corrects a statement of fact. G6, briefs are useful to a person, counts approvals without one.

Frontier model

One of the most capable AI models, offered by an AI company as a hosted service.

In this project: the Claude API, model `claude-sonnet-5-5`, configurable. It is reached through Anthropic's client library behind the same model gateway as Agentic AI Systems: a new library in an existing pattern, chosen by a rule stated before the runs. It ran once, on the stated case that the local model failed the adequacy test in two

rounds, with the same replay set and prompts. It verified 5 of 7, short of the test's 6, and the approver chose its set, so `config.json` ships set to frontier. It costs money per call, depends on an outside service and cannot be pinned the same way. The system never sees patient data, so only figures from open, aggregate data reach it.

Investigator

The agent that gathers the evidence and drafts.

In this project: the Investigator is the language model agent that drafts the demand brief through read-only typed tools. It never sets a level. It writes the summary, the expected range, the models' agreement and the watch points; code writes the back-test error, the track record and the bank holidays as fixed lines.

LangGraph

An open-source Python library for building agents as a graph of steps that share state.

In this project: the agents run as a LangGraph graph, a pattern from Agentic AI Systems. The same graph, tools, schema and Reviewer run on either backend.

Language model (LLM)

A model trained on large amounts of text, which writes text in reply to a prompt. LLM, large language model, is the usual short form; the documents say "language model".

In this project: the language model writes wording only: the Investigator's draft, the Reviewer's wording check and each notification's wording. It adds no forecast accuracy; its value is explanation, tested by G6 and the template comparison.

Live run

A run of the agents on a real model, recorded so it can be replayed.

In this project: three recorded sets of the seven replay briefs. Local round 1 stopped at its 840-second budget after 3 of 7; local round 2 followed one bounded tuning round; the frontier model ran once. The notebook replays all three from their recordings.

Local model

A language model that runs on one's own computer, not on a hosted service.

In this project: `nemotron-3.5-lightning:30b-a3b` in Ollama, pinned by digest `e7a64ff15fb1`, a fingerprint of the exact model file, at temperature 0. It is private, free per call and repeatable, and `config.json` switches to it. It verified no brief in either round. In round 1 its drafts broke the text rules on placeholders, units and model names, and a bug in my code resent the first request, so no retry saw what was wrong. In round 2 every draft hit the 8,192-token output cap, which counts its thinking. A retry with thinking turned off is untried.

Medical device (MHRA)

Software can be regulated as a medical device when its maker intends it for a medical purpose. The MHRA (Medicines and Healthcare products Regulatory Agency) is the UK regulator.

In this project: on my reading of the MHRA guidance (MHRA, 2023) the system is not a medical device, because its output is aggregate operational planning and affects no individual's treatment. That reading is kept as an open risk. The words the MHRA lists as signs of a device, such as "Predicts", "Prognosis", "Triage" and "Risk of", are avoided.

Model backend

The service that runs the language model.

In this project: the model backend is `llm_backend` in `config.json`: `frontier` (the Claude API), the shipped default because the frontier set is the one approved, or `local` (Ollama on this machine). The recordings replay the notebook on either setting. Each brief records the backend that wrote it.

"Not verified, do not act"

A warning mark on a brief that failed its checks.

In this project: a brief that fails the Reviewer twice, on the figure check or the wording check, or an edit that fails the code figure check, reaches the manager with this mark and the failing figures marked. It can never be approved, and it counts as an event for G4, no unchecked number reaches a manager as approved.

Ollama

Free software that runs language models on one's own computer.

In this project: it runs the local model. Recordings of the model calls of every completed replay run let the replays run without it.

Outbox

A file where messages are written instead of being sent.

In this project: notifications go to an outbox file and are never sent.

Output cap

The most tokens one model reply may generate.

In this project: 8,192 tokens for the local model, counting its thinking. Every round 2 local draft reached it, so none could be read.

Owners by role

Named job roles, each accountable for one control.

In this project: the duty manager approves briefs; the board's unscheduled-care lead owns the thresholds and approves each latch reset; the analytics lead runs the re-selection and owns model changes; a clinical safety reviewer signs off before any live use. In the spot check I play the approver, the unscheduled-care lead and the analytics lead, and I say so.

Placeholder

A marked gap in a text that is filled in later.

In this project: the Investigator writes each figure as a placeholder, such as {F1}, and code writes in the value with a label saying what it is, such as "1,120 attendances (SA-MA forecast for w/e 27 Jun 2021)". The words around the placeholders name no number, model, week, hospital or level.

Planted figure

A wrong number put in on purpose to test a check.

In this project: the designed failure. Code plants a figure no tool returned in the Investigator's first readable draft of a second run of the demand week's brief, and the code figure check must catch it at that figure. On the frontier set it did.

Pydantic

A Python library that checks data against a declared structure, a schema.

In this project: every tool's input and output, and the demand brief, are Pydantic schemas. The brief's schema copies the level and has no field for the stated chance. Its validators refuse a draft that writes a figure itself in place of a placeholder.

ReAct

A published pattern in which an agent reasons and calls tools in turn (Yao et al., 2023).

In this project: the Investigator follows its tool-use pattern, but its tools are read-only and it never sets the level.

Replay

Running a past case again from a saved record, so the result can be checked.

In this project: every model call of a completed replay run is recorded, so each replay runs again without the model. "The replay set" is the fixed set of past weeks the replays use.

Reviewer

The step that checks a draft before a person sees it.

In this project: the Reviewer is the code figure check first, then a screen for the words the MHRA lists as indicative of a medical device (MHRA, 2023, p. 20), then a language model wording check. The wording check allows no claim of cause without a tool basis, no figure described as something its label says it is not, and nothing that reads as a clinical instruction. A failure sends the brief back once. The Reviewer is my own design, not a published pattern.

Second reader (Reader B)

A second person who checks the work independently.

In this project: the second reader, Reader B, is a person who did not build the system and who read with an AI assistant, iterating with it on their own feedback. Reader B read four replay briefs, fixed before anyone read a brief, against the six approval criteria without seeing the approver's decisions, and rated the two template pairs. The four are Ninewells' demand brief (w/e 13 Jun 2021), the Royal Infirmary of Edinburgh's surge brief (w/e 6 Oct 2019), the national latch brief and Ninewells' latch brief (both w/e 22 Mar 2020). Reader B made the same decision as the approver on all four, so G6's second half is met.

Stated purpose

What a system's maker says it is for, which also limits how it may be used.

In this project: aggregate operational planning, a forecast of attendances for front-door staffing. It must never rank hospitals, justify cuts or monitor the 4-hour standard. The purpose is stated in the README, the notebook and the case study. Each brief states what the level means and that it is not pressure, need or priority.

Tool

A function an agent can call to get data or act.

In this project: five read-only typed tools: `get_forecast`, `get_history`, `get_backtest_error`, `get_track_record` and `get_calendar` (Scottish bank holidays). Each takes `as_of` and returns only data released by then. The forecast tool never returns the GRU's forecast.

Wording check

A check of how a text describes its facts.

In this project: the Reviewer's last stage, a language model check that runs after the code figure check. It refused both frontier demand-week briefs for claims the brief did not tie to a cited tool result. One was that recent weeks were rising, which the history tool showed but the brief did not cite; the other, that the forecasts fit them, echoed a line code gave the agent.

6. Project terms

Additional information (status)

Information offered beside an existing process, which the process can use but does not depend on.

In this project: the system's status at this stage. It gives the duty manager additional information beside their existing process, replaces no current process and sets no staffing on its own. That is my choice for this stage, not

a shortfall. A primary operational tool needs shadow mode, a pilot with managers who did not build it and tuning over time.

Artefact

The thing built and submitted.

In this project: the integrated artefact is the new system, framework or applied case study built for the Synthesis. Here it is a design case study supported by one spot-check notebook and `requirements.txt`.

Case study

A document that sets out a design and its reasoning.

In this project: the artefact's document, `ae_escalation_case_study`, written after the notebook so it can quote real results. It has fourteen sections, from Overview to the deployment path and next steps.

Data test

The first measurement of the data and the forecast models.

In this project: run on 24 October 2026 before the design's rules. Its back-test protocol was fixed before any result, and the notebook reproduces its figures in section 3's first tables.

Deck

A set of presentation slides.

In this project: the PowerPoint slides planned for the Defence, 14 slides in 13.5 minutes with 1.5 minutes in reserve, covering the seven required sections. They are not yet built.

Defence

The live presentation and questioning that follows the Synthesis.

In this project: Professional Industry Defense, the brief's spelling in the name, "the Defence" for short. It is a 30-minute session with the mentor: a 15-minute presentation, then a 15-minute live defence.

Evidence labels

Labels that say how strong the evidence for a claim is.

In this project: each link of the value chain carries one. Measured: our own in-sample counts (link 1). Sourced: a hypothesised benefit that a checked publication supports, not a demonstrated effect of this system (links 2 to 4). To be tested: the pilot's goal (link 5), also a hypothesis.

Goals G1 to G8

The system's goals, each with a measure and a pass mark, stated before the evaluation runs.

In this project: each is reported as met or not met. G1, G3, G7 and G8 were set by design tuning in the probes, so they are in-sample. G2, G4 and G6 are the real tests. No threshold moves from the notebook's evaluation run onwards.

Goal	What it asks	Met if	Result in the notebook, in-sample
G1	Forecasts a manager can plan with	The primary model within 1 point of the best simple model at every release horizon	Met: 0.18 to 0.74 points below it, measurable only at release horizon 4
G2	Honest uncertainty	90% to 98% of national actuals inside the 95% range	Met: 95.7% to 96.4% inside
G3	Shocks are caught	Alarm known by Tue 31 Mar 2020; at most 1 national false alarm a year	Met: known Tue 31 Mar 2020
G4	No unchecked number reaches a manager as approved	Every approved brief passes the code figure check; the planted figure is rejected	Met: 5 of 5 pass again; no not-verified brief approved; the planted figure caught

Goal	What it asks	Met if	Result in the notebook, in-sample
G5	The agent never sets a level	Enforced by the schema and tested; reported as a safety invariant	Holds: 7 of 7 briefs written
G6	Briefs are useful to a person	All but one approved without a factual edit; the second reader agrees on at least three of four	Not met: 5 of 7 approved without a factual edit, 6 needed, the two short being the refused demand-week briefs; Reader B agreed on 4 of 4
G7	Raised levels are rare in steady years	Under 10% of outlooks raised in every year except 2020 to 2022, with no floor	Narrowly missed: 10.4%, 10.9% and 11.7% in 2018 (from July), 2023 and 2024; not tuned; my untested reading is growth past the years that set capacity, which shadow mode tests
G8	A flag tells a manager something	Outlook flags borne out at least twice as often as the outlook base rate	Met: 2.54 times in the steady years, 3.67 in all years

Integration

Combining parts into one working system.

In this project: integration is combining methods, insights or components from the prior capstone projects into one system. Four are integrated, each doing one job in the weekly flow.

Manager parameter

A setting a manager can turn on for a known factor.

In this project: a manager parameter is a labelled, recorded adjustment for a factor known in advance but one-off, like an authorised forecast adjustment in ERP demand planning. It is a next step, not in this version.

Mentor

The Udacity reviewer.

In this project: the mentor is the Udacity reviewer who hears the presentation and leads the defence.

Model input

A measured factor fed into a forecast.

In this project: a next step. A factor the planners use that recurs with a measured history becomes a model input, but only after it is tested on the back-test.

Next steps

The work that follows this version, in order.

In this project: future pre-deployment investigations after this base is proven. The case study's section 14 sets out three, in order, as the path from additional information to a primary operational tool:

1. Shadow mode before any go-live. First I close two gaps this build found: a trend claim that cites one week, and a trial of the local model with its thinking turned off. A third, the drift line's claim that the forecasts fit recent weeks, is closed in the shipped template and in new runs. The template rule's model-disagreement ground also needs a threshold.
2. A pilot that measures throughput, with managers who did not build the system and bands set per hospital for fairness. Within it I learn what planners use and route each factor to a model input, a manager parameter or their judgement; weather, population demographics, and flu strains with their expected severity are named candidates. Tuning over time then acts on what shadow mode and the pilot find.
3. The wider system: once the base is proven, admissions and occupancy data, so the same pattern can speak to beds and social care.

The paper's Figure 8 draws the path to a primary operational tool: shadow mode, then a pilot with managers who did not build it, then tuning over time on their feedback and on what planners use.

Paper

The written report for the Synthesis.

In this project: the Reflective Synthesis Paper, file `module_summary.pdf`, "the paper" for short. It has the brief's seven sections and a body of 1,500 to 2,000 words of prose, written in the APA template in APA 7 style. Its tables, figures, figure notes and references are not counted.

Pilot

A small trial in real use, to measure an effect before wider rollout.

In this project: this version is the pilot build, kept a pilot design on purpose: it gives additional information only and has had no real use. The live pilot, a pilot with managers who did not build the system and with bands set per hospital for fairness, would measure A&E throughput, link 5 of the value chain. It comes after shadow mode.

Prior capstone project

An earlier project in the same programme.

In this project: a prior capstone project is one of the six earlier capstone projects, named, never numbered. Four are integrated:

- AI Programming Foundations: load, clean, flag and keep.
- Conduct a Statistical Analysis Using Python, "Statistical Analysis" after first use: intervals, back-test and comparison.
- Deep Learning Systems: the GRU comparator.
- Agentic AI Systems: agents, tools, approval queue, replay and audit.

Applied Machine Learning and Generative AI Applications are not integrated. Each is named, never cited as a source.

Probe

A scripted check on the data that answers one design question.

In this project: nine probes on 24 and 25 October 2026, on CPU, with no language model. Probe 5 chose the design and probe 6 the flag's 50% setting, each by a rule written before its results. Probe 7 measured the approved design against the goals with nothing tuned, probe 8 counted the notice it gives, and probe 9 tested repeated flags. All are in-sample.

Shadow mode

Running a finished system on live data without acting on its output, to test it before go-live.

In this project: shadow mode is the frozen design run on new weekly releases, with every brief written and logged but none acted on, before any go-live. It is never called a "shadow run". It scores the goals on weeks the design has never seen. It is a pre-deploy step, not done before submission.

Spot-check notebook

A notebook that runs part of a design on real data, to show observed behaviour.

In this project: `ae_escalation_spot_check.ipynb`, run end to end on the PHS data. It covers the data, forecasts, back-test, level and drift rules, the agents and approvals on the replay set, and the evaluation. It replays every recorded model call of the completed replay runs, so it needs no language model or GPU.

Synthesis

The final integrating project of the capstone.

In this project: Industry Integrated AI System Synthesis, "the Synthesis" for short, never numbered.

Throughput

How many patients a department sees and moves on in a given time.

In this project: better A&E throughput is link 5 of the value chain, to be tested in a pilot. This version does not claim it.

Value chain

The steps by which a tool's output could lead to a benefit.

In this project: five links, each with its evidence label. Link 1 is measured in-sample. Links 2 to 4 are hypothesised benefits that the sources support, and link 5 is a hypothesis the pilot would test; none is an effect this system has shown.

Link	What it is	Label	Evidence
1	Earlier notice of unusually busy weeks	Measured, in-sample	Notice of 57.0 of 142.5 busy hospital-weeks in the mean steady year; 280 of 713 over the steady years, 197 by the week's own flag
2	Planned bank cover instead of late agency cover	Sourced	Audit Scotland (2025); Public Services Delivery Scotland (2026); Hu et al. (2025), one US ED, before and after, no control group
3	Less stress for permanent staff	Sourced	Imamura et al. (2026); RCEM (2026)
4	Better patient care	Sourced	Rivet et al. (2026) and Jones et al. (2022), associations, not proof
5	Better A&E throughput	To be tested	The pilot's goal

References

Audit Scotland. (2025, December 4). *NHS in Scotland 2025: Finance and performance*. https://audit.scot/uploads/2025-11/nr_251204_NHS_in_Scotland_2025.pdf

European Union. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), as amended by Regulation (EU) 2026/1744. *Official Journal of the European Union*, L, 2024/1689. http://data.europa.eu/eli/reg/2024/1689/oj

Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association, 19*(1), 121-127. https://doi.org/10.1136/amiajnl-2011-000089

Hu, Y., Chan, C. W., Dong, J., Kazekjian, A., Ophaswongse, C., Sugalski, G., Underwood, J. P., & Perotte, R. (2025). Implementing a prediction driven framework for emergency department nurse staffing to optimize real time decisions. *npj Health Systems, 2*(1), Article 16. https://doi.org/10.1038/s44401-025-00019-2

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. https://otexts.com/fpp3/

Imamura, Y., Sakuramoto, H., Ouchi, A., Honda, T., Uno, S., & Tamoto, M. (2026). Emergency nurse staffing and outcomes for patients and nurses: A scoping review. *Journal of Emergency Nursing*. Advance online publication. https://doi.org/10.1016/j.jen.2026.05.015

Jones, S., Moulton, C., Swift, S., Molyneux, P., Black, S., Mason, N., Oakley, R., & Mann, C. (2022). Association between delays to patient admission from the emergency department and all-cause 30-day mortality. *Emergency Medicine Journal, 39*(3), 168-173. https://doi.org/10.1136/emermed-2021-211572

Medicines and Healthcare products Regulatory Agency. (2023). *Medical device stand-alone software including apps (including IVDMDs)* (Version 1.10g; file revised October 2026). GOV.UK. Retrieved October 24, 2026, from https://assets.publishing.service.gov.uk/media/6abf6c8c943e890cb55e5e52/Medical_device_stand-alone_software_including_apps__including_IVDMDs.pdf

NHS England. (2024, December 2). *Integrated operational pressures escalation levels (OPEL) framework 2024 to 2026* (page updated 23 May 2025). https://www.england.nhs.uk/publication/integrated-opel-framework-2024-to-2026/

NHS England. (2025). *Resident Doctors (RDs): Rota Code of Practice compliance data collection* (DCB3042 AMD 35/2025) [Data collection]. https://standards.nhs.uk/published-standards/resident-doctors-rds-rota-code-of-practice-compliance-data-collection

Public Health Scotland. (2026a). *Weekly A&E activity and waiting times* [Data set]. Scottish Health and Social Care Open Data. Retrieved October 24, 2026, from https://www.opendata.nhs.scot/dataset/weekly-accident-and-emergency-activity-and-waiting-times

Public Health Scotland. (2026b, October 6). *Accident and emergency: Main points, all A&E sites*. Retrieved October 27, 2026, from https://publichealthscotland.scot/healthcare-system/urgent-and-unscheduled-care/accident-and-emergency/main-points/all-ae-sites/

Public Services Delivery Scotland. (2026, June 2). *NHSScotland workforce: Latest statistics at 31 March 2026* [Accredited official statistics]. Turas Data Intelligence. https://turasdata.nes.nhs.scot/data-and-reports/official-workforce-statistics/all-official-statistics-publications/02-june-2026-workforce/?pageid=16251

Rivet, C., Amazouz, H., Gratacap, M., Delannoy, Q., Yordanov, Y., Leblanc, J., & URGE collaborators. (2026). Association between patient-to-nurse ratios in emergency departments and patient, staff, and organizational outcomes: A systematic review. *European Journal of Emergency Medicine, 33*(4), 237-247. https://doi.org/10.1097/MEJ.0000000000001350

Royal College of Emergency Medicine. (2026). *The state of emergency medicine in Scotland* [Report]. https://rcem.ac.uk/wp-content/uploads/2026/03/RCEM_SoEM_Scotland_A4_0326.pdf

Scottish Government. (2016). *Emergency department capacity management: Guidance*. https://www.gov.scot/publications/emergency-department-capacity-management-guidance-2/

Scottish Government. (2025, June 30). *NHS Greater Glasgow and Clyde emergency department review: Response*. https://www.gov.scot/publications/nhs-greater-glasgow-and-clyde-emergency-department-review-response/pages/response/

Scottish Government. (2026, October 8). *Health and social care - surge preparedness: National planning priorities and principles*. https://www.gov.scot/publications/health-social-care-surge-preparedness-national-planning-priorities-principles/

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=WE_vluYUL-X